James Harris

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TECHNICAL EXPERIENCE

Quantum Testing Framework September. 2025 - May. 2026
Probabilistic Testing Framework for Quantum Simulators (Python, Qiskit, Pytest) Cardiff, UK

- Achieved a scalable bridge between classical engineering and quantum research by integrating the testing framework with the Pytest lifecycle, enabling automated verification of quantum properties like Superposition and Entanglement within modern Continuous Integration pipelines.
- Contributed to solving the Test Oracle Problem in quantum computing by replacing deterministic binary pass/fail logic with Threshold Assertions, which allow developers to define statistical tolerance parameters and confidence intervals for probabilistic quantum state distributions.
- Engineered a Probabilistic Validation Framework by developing a high-level Python wrapper that abstracts the quantum execution lifecycle including transpilation, simulation, and sampling into a modular architecture compatible with standard software engineering practices.

Parallel Systems Simulation Comparison (Langton's Ant) Oct. 2025 - May. 2026
Langton's Ant Cellular Automaton (Java, JavaFX, Multi-threading, Parallelization) Cardiff, UK

- Engineered a high-concurrency simulation engine using the Strategy Design Pattern, achieving a peak 7.26x speedup by transitioning from sequential execution to a coarse-grained ExecutorService.
- Optimized memory efficiency by implementing a sparse grid via ConcurrentHashMap and bit-shifting for 64-bit coordinate encoding, reducing space complexity from $O(n^2)$ to $O(K)$, where K represents the active cell count.
- Analyzed three distinct concurrency models: Fine-Grained ExecutorService (identifying a 2.3x performance degradation due to task overhead), Coarse-Grained Batching (7.26x speedup using Recursive Decomposition), and ForkJoinPool (7.18x speedup using Work-Stealing).

Product Cipher in Feistel Network January. 2026 - May. 2026
Symmetric Block Cipher Implementation (C++, Cryptography, TDD) Cardiff, UK

- Developed a 16-round symmetric block cipher based on a Feistel Network architecture using C++, implementing a complex round function that integrates Caesar, Vigenère, and Columnar Transposition ciphers to achieve high levels of confusion and diffusion by layering substitution and transposition techniques.
- Established an efficient validation pipeline using Test-Driven Development to ensure 100% functional accuracy of the encryption/decryption cycles and verified the avalanche effect through automated unit tests.

EDUCATION

Cardiff Metropolitan University September. 2023 – May. 2026
BSc (Hons) Software Engineering First Class Honours Cardiff, UK

- **Achievement**: Awarded Software Engineering Student of the Year for Best Overall Performance
- **Accreditation**: BCS and IET accredited programme (Partial CEng)
- **Modules**: Networks and Security, Cyber Security and Cryptography, Object Oriented Programming, Algorithms and Data Structures, Agile Development and DevOps, Parallel and Distributed Systems

Cardiff and Vale College September. 2020 – May. 2022
Foundation Degree Graphic Communication (Merit) Cardiff, UK

- **Modules**: Collaborative Working, Entrepreneurial Enterprise
- **Student Representative**: Represented students in meetings with faculty, academic staff and college management by communicating feedback, concerns, and suggestions to support improvements to both the student and teacher experience.

AWARDS

Software Engineering Student of the Year 2026
Awarded by Cardiff Metropolitan University Cardiff, UK

- **Achievement**: Awarded for outstanding academic performance and overall achievement throughout the Software Engineering programme.

First Class Honours BSc Software Engineering 2026
Awarded by Cardiff Metropolitan University Cardiff, UK

- **Achievement**: Achieved the highest undergraduate classification, demonstrating excellent performance across software engineering principles, development practices, and technical projects.

Progression Award 2024/25
Awarded by Cardiff Metropolitan University Cardiff, UK

- **Achievement**: Awarded in recognition of successful academic progression and continued achievement throughout the Software Engineering programme.

SKILLS

Languages: Python, C++, Java, SQL, Dart
Developer Tools: Git, Docker, Pytest, Qiskit, JUnit, Flutter, JavaFX
Methodologies: Test-Driven Development, Agile Development, Behaviour Driven Development

WORK EXPERIENCE

Fisherman's Rest Inn July 2021 – September 2024
Supervisor / Front of House Bedwas, UK

- Supervisor (2024): Led shifts, trained new staff, resolved customer complaints and concerns, managed daily operations such as coordinating staff roles and filling in for missing team members.
- Front of House (2021 – 2022): Worked as a bartender, serving drinks and handling payments. Took delivery orders and bookings over the phone, managed reservations, and provided customer service. Opened and closed the pub, including setting up for service and completing end-of-day tasks.